

QOptiSolve — Technical Paper

Abstract

QOptiSolve is a quantum optimization solver that applies the **Quantum Approximate Optimization Algorithm (QAOA)** to combinatorial problems such as portfolio allocation and graph Max-Cut. It compares quantum outputs with classical solvers, providing visualizations of convergence and solution quality. Built with a modular design and a test-driven development (TDD) approach, QOptiSolve demonstrates how quantum algorithms can be integrated into optimization workflows and analyzed alongside classical methods.

1. Introduction

Optimization under uncertainty is a fundamental challenge in fields ranging from finance to logistics. Quantum algorithms such as QAOA have been proposed as potential accelerators for solving NP-hard problems. **QOptiSolve** provides a reproducible, educational framework to explore quantum optimization, benchmark results against classical solvers, and visualize solution landscapes.

2. System Architecture

QOptiSolve is structured as a modular pipeline:

1. **Problem Definition:** Users specify optimization problems (e.g., Max-Cut graphs, portfolio models).
 2. **QAOA Module:** Builds parameterized quantum circuits for problem Hamiltonians.
 3. **Classical Module:** Solves the same problem with deterministic or heuristic algorithms.
 4. **Comparison Engine:** Evaluates quantum vs. classical results on cost and optimality.
 5. **Visualizer:** Displays convergence plots, cost landscapes, and allocation/cut outputs.
 6. **UI Layer:** Streamlit interface for interactive problem setup and solver execution.
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3. Methods and Algorithms

3.1 Quantum Approximate Optimization Algorithm (QAOA)

- Problem encoded as an Ising Hamiltonian.
- Alternating application of cost Hamiltonian and mixer Hamiltonian.
- Optimized via classical parameter tuning (variational approach).
- Depth parameter (p) controls trade-off between accuracy and circuit complexity.

3.2 Classical Optimization Methods

- **Graph Problems:** Greedy Max-Cut, semidefinite programming approximations.

- **Portfolio Problems:** Mean-variance optimization, risk-constrained solvers.

3.3 Comparative Metrics

- **Cut Value** (graph problems).
 - **Expected Return and Risk** (portfolio optimization).
 - **Approximation Ratio:** Quantum vs. optimal classical.
 - **Convergence Behavior:** Optimization trajectory of variational parameters.
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4. Implementation

- **Language:** Python.
 - **Libraries:** Qiskit for quantum circuits, NumPy/Pandas for data handling, Streamlit for visualization.
 - **Design Pattern:** Clear separation between problem encoding, quantum solvers, classical solvers, and visualization.
 - **Deployment:** Local execution with Qiskit Aer simulator; optional cloud backends (IBM Q, Amazon Braket).
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5. Evaluation

5.1 Datasets

- Synthetic graphs (5–20 nodes) for Max-Cut experiments.
- Simulated asset returns and covariance matrices for portfolio optimization.

5.2 Metrics

- **Optimality Gap:** $|\text{Quantum} - \text{Classical Optimal}|$.
- **Convergence Rate:** Iterations to reach stable parameters.
- **Runtime Performance:** Simulation times for increasing depth (p).

5.3 Results (Pilot)

- Max-Cut experiments show QAOA ($p=2$) achieving ~95% of classical optimal solutions.
 - Portfolio optimization: QAOA approximates Kelly-weighted allocations with higher variance.
 - Runtime grows exponentially with depth, highlighting scalability limits.
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6. Testing Strategy

QOptiSolve employs TDD practices: - **Unit Tests:** Quantum circuit generation, classical solver correctness. - **Integration Tests:** End-to-end pipeline from problem definition → solution comparison. - **Mock Backend Tests:** Deterministic results with simulator seeds. - **UI Tests:** Streamlit dashboard rendering.

7. Discussion

QOptiSolve demonstrates both the promise and limitations of near-term quantum algorithms. While QAOA can approximate solutions competitively at low depths, scalability and noise remain challenges. Comparative analysis ensures transparency by showing quantum performance relative to classical baselines.

8. Future Work

- Extend problem library (scheduling, traveling salesman, knapsack).
 - Implement hybrid quantum-classical optimization loops.
 - Integrate cloud quantum hardware for real QPU experiments.
 - Add performance benchmarks across simulators.
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9. Conclusion

QOptiSolve provides a modular, test-driven environment for exploring quantum optimization. It enables side-by-side comparisons of quantum and classical solvers, making it a practical educational tool for understanding the evolving landscape of quantum algorithms.

References

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